CRF State Summary v4 (Final)

Maturity 8.0/8 (Theoretical) · 7.5/8 (Empirical)

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1 The Chain

$$\delta \to \mathcal{P} \to \mathcal{C} \to D_{\text{KL}} \to \theta \to \mathcal{R} \to t$$

2 Results Table

Result	**Status**	**Source**
Core Physics		
$n = 3$ from SO(3)	Derived	#16
Born Rule via Gleason	Theorem	Session 11
Schwarzschild, $G = \ln 2 \cdot c^2/D_0$	Derived	Paper v4
$T_{\mu\nu}$, EIC Bridge, $I_{\text{eff}} \approx \ln 2$	Derived	EIC, Track B.1
Instability #21 — Full Chain		
$K^* = 0.1170$, $a = (n-1)/2 = 3.5$	Derived	θ-formula
$T_{\text{avg}} = 2nK^*/[(n-1)\epsilon_0] = 35.4$	**DERIVED**	**#21a, 1.2%**
$H_{\text{before}} = 1.764 \pm 0.15$	**MEASURED**	**#21b**
$H_{\text{uniform}} = 2.055$, $H_{\text{peaked}} = 0.304$	Derived	digamma
$f_{II} = 0.147$, $H_{\text{after}} = 1.798$	Measured	V20.3
$F = 1.527$	Measured	V20.3
$\beta_{\text{derived}} = 2.219$	**DERIVED**	**#21 CLOSED 1.7%**
$\beta = d_s/(2\ln 2)$ formally	Open at 1.5%	Algebraic identity
Node Lifecycle		
F_i: $81 \to 1.43$ (#HN-01)	**Confirmed**	V20.3
HN-02 aging	Not triggered	64k sweeps
HN-03 collective excess	Open	Cluster tracking
HN-04 hysteresis	**Derived**	SO(3) Thm 1
HN-05 $\Omega = \kappa N_{\text{eff}}$	**Derived**	By construction
$f_{II} = 0.147$ (Phase II fraction)	Measured	Connects to HN-03
P0-H Empirical		
3 subjects, inverted-U accuracy(RT)	Confirmed	v5, 80 trials
$\omega = 9.29$ rad/s, $t_1 \approx 300$ms, $t_2 \approx 485$ms	Measured	P0-H v5
3-type classifier (visual/somatic/blocked)	Predicted	Testable
UF–CRF bridge, barami formalism	Derived	UF Addendum
Other		
Resolution scale $R_0, R_1, R_{\max}$; collapse premium ≈ 19	Derived	This session

3 Key Numbers

$\alpha = \ln 2$	$\epsilon_0 = 0.00755$	$D_0 = 0.9403$	$\kappa = 0.15$
$K^* = 0.1170$	$d_s = 3.031 \pm 0.076$	$\beta = 2.212 \pm 0.015$	$F = 1.527$
$H_{\text{before}} = 1.764$	$H_{\text{after}} = 1.798$	$T_{\text{avg}} = 35.4$	$f_{II} = 0.147$
$t_1 \approx 300\text{ms}$	$t_2 \approx 485\text{ms}$	$\omega = 9.29$ rad/s	

4 Open Gaps

Gap	Priority	Path
$\beta = d_s/(2\ln 2)$ algebraic	1	Show formula $= d_s/(2\ln 2)$ analytically
f_{II} from lifecycle	2	HN-03: Phase II fraction from $\mathrm{T}_avg + firingrate$
P0-H pre-registration	1	$n \geq 50$, Subject 4 (blocked type)
P0 SNSPD	2	Lab access; $\tau_1 < \tau_2$
HN-02 aging	3	64k sweeps
Kerr metric	4	Rotating mass

Maturity: 8.0/8 Theoretical · Gate to Full 8: Empirical

Theoretical chain from δ to β is complete at 1.7%.
Remaining theoretical: algebraic proof of $\beta = d_s/(2\ln 2)$ (1.5% gap).

Gate to Empirical Level 8:
(A) P0-H $n \geq 50$ + Subject 4 (blocked type, flat 50%)
(B) P0 SNSPD $\tau_1 < \tau_2$ at $> 5\sigma$
Either gate constitutes first experimental contact with the framework.

5 Documents This Session

CRF_21a_Closed · CRF_21b_Closed · CRF_21_Closed (1.7%) · CRF_21_F_Measured · CRF_21_Full · CRF_21_Boundary · CRF_21a_Kramers · CRF_NodeLifecycle_v3 · CRF_P0H_3Subjects · CRF_P0H_InvertedU · CRF_P0H_DoublePeak · CRF_UF_Addendum · CRF_Resolution_Scale